# Ambient IoT Backscatter Sensing for Fall Detection and Localization in Smart Healthcare

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***Abstract*—Falls remain a major cause of injury and death among older adults, which shows the need for reliable and non-intrusive monitoring solutions in healthcare environments. In this paper, we propose a novel Ambient Internet of Things (IoT) backscatter sensing system that utilizes a dense array of passive tags and a minimal number of reader antennas for cost-effective fall detection and localization. To fully exploit the spatial and temporal characteristics of ambient backscatter sensing data, we design a hierarchical multi-task spatio-temporal graph attention network (HM-STGAT), which jointly models the spatial relationships among tags and antennas as well as the temporal dynamics of human activities. The proposed unified framework simultaneously detects fall events and accurately estimates fall locations. We validate the proposed approach through a real-world experiment to collect a diverse dataset of fall and non-fall scenarios. Experimental results demonstrate that the proposed method achieves state-of-the-art performance in both fall detection accuracy and localization precision, highlighting its potential for practical deployment in healthcare monitoring applications.**

***Index Terms*—AI, ambient IoT, backscatter communication, fall detection, healthcare, ISAC, localization, multi-task learning, net-zero, RFID, sensing, spatio-temporal learning.**

## I. Introduction

FALLS are a leading cause of unintentional injury death among older adults and impose a significant burden on healthcare systems worldwide. According to the World Health Organization (WHO), hundreds of thousands of people die from falls every year [1]. Implementing effective fall-prevention strategies can reduce both injury incidence and healthcare costs. In particular, automatic fall detection systems that provide continuous monitoring and timely alerts are of great importance in smart healthcare environments.

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Existing fall detection approaches can be broadly categorized into wearable and non-wearable systems [2]. Wearable methods typically rely on sensors such as accelerometers and gyroscopes embedded in belts, watches, and wristbands [3], [4], [5]. Recent studies have shown that such wearable systems can achieve very high detection accuracies under controlled conditions [6], [7]. However, they depend on correct placement and regular battery maintenance, which are often difficult to guarantee in long-term daily use. Non-wearable methods exploit environmental sensing technologies such as cameras, radars, WiFi, infrared (IR), and other ambient monitoring sensors. Although many of these technologies can achieve accurate fall detection, they suffer from practical limitations in real-world deployment. Vision-based systems raise serious privacy concerns in sensitive areas such as bathrooms and bedrooms [8]. UWB and mmWave radars have relatively high hardware cost, and their performance can be degraded by moisture [9], [10]. WiFi-based sensing is strongly affected by interference from communication traffic [11]. Methods based on infrared sensors, vibration sensing, and acoustic sensing are sensitive to temperature, floor material, and background noise [12], [13], [14]. As a result, there is a growing need for cost-effective, privacy-preserving, and robust fall detection solutions that can be deployed in smart healthcare environments.

Ambient Internet of Things (IoT) technology has recently emerged as a promising solution for large-scale and battery-free healthcare applications [15]. It has also been formally included in the 3GPP Release 19 standardization [16]. Within this vision, radio frequency identification (RFID) is a representative technology. Passive RFID tags are low-cost, battery-free, and non-intrusive. They avoid the privacy concerns associated with vision-based and acoustic sensing methods, which makes them particularly attractive for privacy-sensitive indoor spaces. Moreover, compared with sensing technologies that can be sensitive to moisture, temperature variations, or background noise, RFID backscatter sensing is a promising candidate for deployment in diverse indoor environments. These characteristics motivate the use of RFID backscatter sensing for smart healthcare monitoring within the Ambient IoT framework.

RFID-based fall detection systems can be broadly divided into wearable-tag and contactless-tag approaches. Wearable-tag solutions attach passive or semi-passive tags to the human body to capture motion patterns. For example, RSS fingerprinting has been applied to wearable human activity recognition and indoor localization using body-worn RFID tags [17], and Doppler-shift

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features from passive tags attached to the body have been exploited to distinguish falls from other motions [18]. Custom e-health platforms integrating RFID readers have also been used to perform fall detection in indoor environments, but they still require the user to carry tags [19]. In addition, passive RFID sensor tags attached to footwear have been used for elderly fall detection by leveraging RSSI and on-tag pressure measurements [20]. In contrast, contactless RFID systems place tags in the environment, such as on walls, floors, or furniture. Falls are inferred from the changes in backscattered signals caused by human motion. An early device-free scheme in [21] combined angle-based outlier detection, dynamic time warping, and $k$-nearest neighbors to detect fall events and estimate fall direction using UHF passive tags. More recently, deep learning has been introduced into this setting, including tag-free fall detection based on a transformer encoder [22] and wear-free fall detection using deep residual networks on RFID signal features [23]. Multidirectional fall detection and pseudo-localization have also been demonstrated by deploying RFID tag arrays in the environment [24].

These works clearly demonstrate the potential of RFID for unobtrusive fall detection and even coarse localization. However, most existing RFID-based approaches still suffer from several limitations. Wearable-tag methods require users to carry tags, whereas contactless methods typically treat RSS measurements as independent time series or per-tag feature vectors, without explicitly modeling the underlying spatial structure of the sensing environment. In addition, fall detection and fall localization are often handled separately, rather than being jointly optimized within a unified learning framework. These limitations are closely related to the way RFID sensing data are modeled and processed. From a modeling perspective, spatio-temporal graph neural networks have been widely explored for video-based activity recognition and fall detection [25], [26], [27]. These methods suggest that graph-based spatio-temporal modeling can effectively capture complex human motion patterns. However, to the best of our knowledge, such spatio-temporal graph modeling has not yet been explored for RFID-based fall detection. Moreover, it has not been used to jointly learn fall detection and fall localization within an Ambient IoT backscatter sensing framework.

Motivated by the above observations, we design an Ambient IoT backscatter sensing system that densely deploys passive RFID tags on a vertical wall to form a two-dimensional tag grid. Human motions and falls produce recognizable spatio-temporal patterns in the RSS measured across the tag grid. To fully exploit the spatial relationships among tags and antennas as well as the temporal dynamics of human activities, we develop a hierarchical multi-task spatio-temporal graph attention network (HM-STGAT) operating directly on a graph constructed over tag and antenna nodes. The proposed framework simultaneously outputs a binary fall detection decision and an estimated fall location.

The main contributions of this paper are as follows

- We propose a novel passive RFID tag grid-based healthcare monitoring system that utilizes a dense deployment of passive RFID tags with a minimal number of reader antennas, so as to realize non-intrusive and cost-effective motion sensing, fall detection, and fall localization in indoor healthcare environments. Beyond sensing, the RFID tags also incorporate memory, allowing them to support communication functions and thus offering the potential for integrated sensing and communication (ISAC) applications.
- We design a hierarchical multi-task spatio-temporal graph attention network that effectively captures spatial and temporal relationships of the sensing data for joint fall detection and localization. This unified framework simultaneously achieves high accuracy of fall event detection and precise fall localization. As far as we know, we are the first to propose the use of HM-STGAT for both fall detection and fall localization in Ambient IoT sensing, thereby demonstrating its effectiveness in a unified multi-task setting.
- We propose a joint optimization model for fall detection and localization, incorporating adjustable trade-off factors. By emphasizing the fall detection factor, the model achieves high accuracy in detecting falls. In contrast, when emphasizing the localization factor, the model achieves high localization precision. With a balanced trade-off factor, the model can perform both fall detection and localization simultaneously, while maintaining a controlled compromise between the two tasks.
- We conduct a hardware experiment in an indoor laboratory and build a comprehensive real-world RFID sensing dataset consisting of diverse fall and non-fall scenarios. We show that even with a relatively small dataset, our approach can realize high fall detection accuracy (97.5%) and precise localization performance, validating the practicality of Ambient IoT-enabled healthcare monitoring.

The remainder of this paper is structured as follows: Section II presents the detailed system model. Section III introduces our proposed HM-STGAT methodology. Section IV provides the performance analysis. Section V describes the experimental setup and evaluation metrics. Section VI presents measurement results and comparative analysis. Finally, Section VII demonstrates the conclusion of this paper.

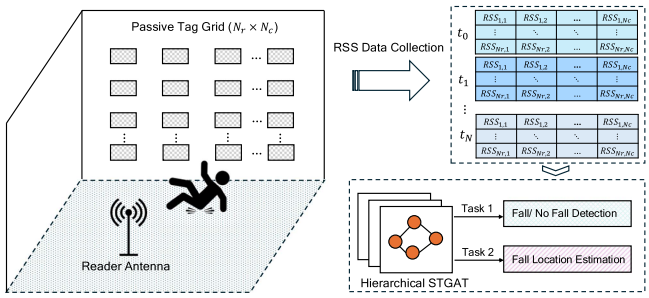



Fig. 1. System model of the RFID-based fall detection system.

## II. System Model

As shown in Fig. 1, we deploy a spatial grid composed of passive RFID tags $N_r$ rows and $N_c$ columns, forming a $N_r \times N_c$

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tag grid on a vertical wall. This structured spatial configuration enables the capture of received signal strength (RSS) variations arising from human movements within the monitored area. In front of the tag grid, we place $N_{\text{ant}}$ distributed reader antennas at strategic positions to monitor the environment. These antennas continuously collect RSS data via backscatter communication between the reader antennas and passive RFID tags.

### A. Backscatter Communication Mechanism

Theoretically, the RSS obtained at a reader antenna depends mainly on transmitted power $P_T$, antenna gain, reader's operation frequency $f_r$, distance $d_i^{(m)}$ between antenna $m$ and tag $i$, and environmental factors such as multipath effects and human presence. For a monostatic RFID system that uses the same antenna for transmission and reception, the RSS received by antenna $m$ from tag $i$ can be modeled as [28]

$$\text{RSS}_i^{(m)} = P_T G_{\text{ant}}^2 G_{\text{tag}}^2 \left(\frac{\lambda}{4\pi d_i^{(m)}}\right)^4 F_{\text{mod}}, \tag{1}$$

where $G_{\text{ant}}$ and $G_{\text{tag}}$ represent the gains of the reader antenna and tag, respectively, $\lambda$ is the wavelength corresponding to operation frequency $f_r$, and $F_{\text{mod}}$ is a factor representing modulation efficiency and environmental effects.

Conventionally, RFID tags are equipped with memory that enables basic communication functions. Since this is an inherent capability of standard RFID tags, their communication performance is not specifically analyzed in this work.

In practical scenarios, commercial RFID systems also provide RSS values as received signal strength indicator readings, which are extracted directly from reader hardware. These RSSI measurements vary with reader-tag distances and environmental disturbances such as human movements, which makes it convenient for sensing tasks such as fall detection.

### B. RSS-Based Sensing Signal for Fall Detection

RSS measurements exhibit sensitivity to environmental changes, particularly those brought by human activity. This property makes RSS measurements suitable for monitoring and detecting falls within indoor environments. Our approach utilizes these variations by calculating the deviation of current RSS measurements from a baseline RSS that is obtained under static and interference-free conditions.

Specifically, we define RSS variation $\Delta\text{RSS}$ for each tag $i$ captured by antenna $m$ at a given time step as follows

$$\Delta\text{RSS}_i = \text{RSS}_{\text{Baseline}} - \text{RSS}_{\text{Current},i}^{(m)}, \tag{2}$$

where $\text{RSS}_{\text{Baseline}}$ represents the baseline RSS value, computed by averaging multiple RSS measurements taken over a period of no human presence or significant environmental disturbance. Conversely, $\text{RSS}_{\text{Current},i}^{(m)}$ denotes the current RSS reading obtained during active periods, including scenarios such as standing, walking, or falling near certain tags.

By sequentially collecting these RSS variations across a sequence of time steps, we construct a spatio-temporal RSS variation matrix for each antenna, represented by [29], [30]

$$\Delta\mathbf{RSS}^{(m,t)} = \underbrace{\begin{pmatrix} \Delta\text{RSS}_{1,1}^{(m,t)} & \cdots & \Delta\text{RSS}_{1,N_c}^{(m,t)} \\ \vdots & \ddots & \vdots \\ \Delta\text{RSS}_{N_r,1}^{(m,t)} & \cdots & \Delta\text{RSS}_{N_r,N_c}^{(m,t)} \end{pmatrix}}_{N_r \text{rows} \times N_c \text{columns}}, \tag{3}$$

where $\Delta\text{RSS}_{p,q}^{(m,t)}$ denotes the RSS variation observed at antenna $m$ from tag located at the $p$-th rows and the $q$-th column, at time step $t$. This matrix captures both the spatial distribution and temporal evolution of RSS fluctuations, which creates a robust and data-rich foundation for fall detection analysis.

Importantly, from a temporal perspective, a fall usually leads to a short duration and high magnitude transition in $\Delta\mathbf{RSS}^{(m,t)}$. From a spatial perspective, the human body affects multiple nearby tag-to-antenna links simultaneously. Therefore, the constructed spatio-temporal $\Delta\mathbf{RSS}$ matrices contain temporal abruptness and spatial correlation cues, which can be exploited by a learning model for fall detection and localization.

### C. RFID Tag Grid-Based Fall Localization

Traditional RFID-based indoor localization schemes typically rely on triangulation setups involving at least three spatially separated antennas. Our proposed approach significantly simplifies this requirement by utilizing a dense and structured grid of inexpensive passive RFID tags rather than numerous costly reader antennas. Given that reader antennas are significantly more expensive than passive RFID tags, using a dense tag grid substantially lowers the overall system cost and simplifies deployment and maintenance.

In our fall localization scheme, the dense spatial arrangement of the tags guarantees extensive sensing coverage. Each reader antenna communicates simultaneously with numerous tags, creating a rich mesh of RF links. When a person falls, their body brings fluctuations in the RSS across adjacent tags, forming distinct interference patterns. By analyzing these RSS variations throughout the structured tag grid on a vertical wall, our system can estimate the fall location without needing extensive antenna deployments. For example, we can feed these RSS variations across the entire grid into a graph neural network (GNN), which learns to map these interference patterns directly to precise fall coordinates. This grid-based architecture not only reduces costs but also improves scalability and adaptability, making it highly suitable for diverse real-world indoor environments.

## III. Hierarchical MultiTask SpatioTemporal Graph Attention Network

In this section, we propose an HM-STGAT method for fall event detection and location estimation simultaneously. Specifically, the proposed HM-STGAT comprises three main modules: spatial module, temporal module, and multi-task module, as shown in Fig. 2.

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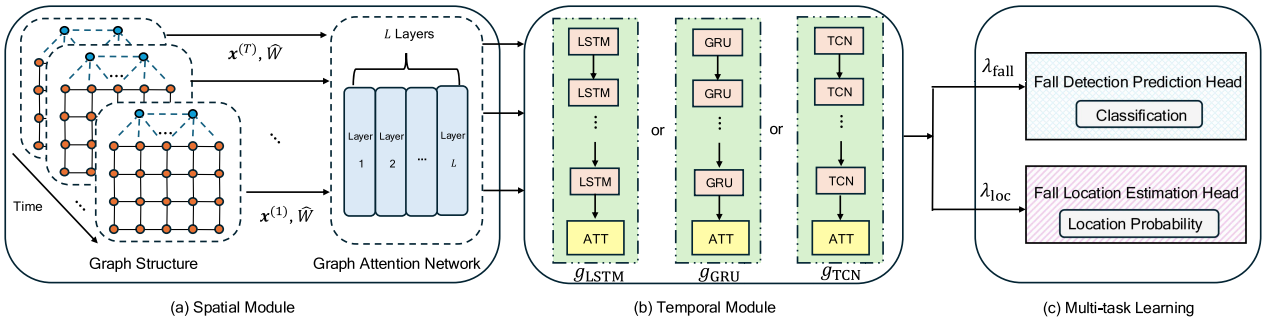



Fig. 2. Diagram of the proposed HM-STGAT model.

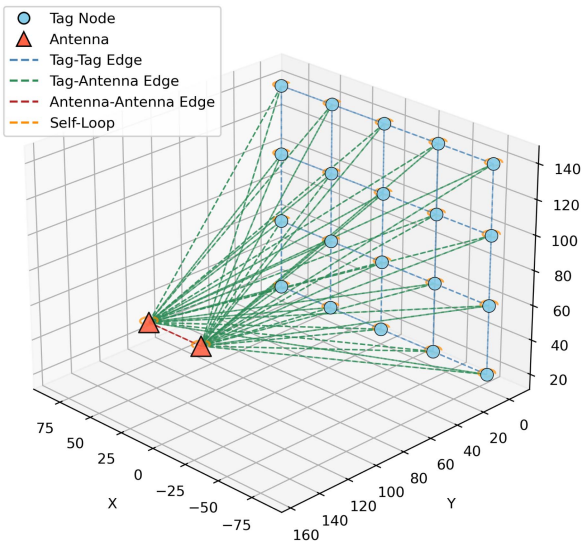



Fig. 3. Hierarchical 3D graph structure.

### A. Spatial Module

Since the location of the tags and the reader antennas can both have a significant effect on the fall detection performance, we model them together in a two-tier spatial graph. Different from the traditional graph structure that treats all the nodes and edges as the same level, we consider a twotier graph structure due to the distance difference of tags and antennas. We consider one tier as local tagtotag edges and one tier as a longrange tagtoantenna and antennatoantenna edges. Antenna nodes can gather and redistribute global information, while tag nodes focus on local neighborhood interactions.

*1) Graph Structure:* At each time step, we build an undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. As shown in Fig. 3, it contains two types of nodes, i.e., tag nodes and antenna nodes, which are defined by

$$\mathcal{V} = \mathcal{V}_{\text{tag}} \cup \mathcal{V}_{\text{ant}}, \tag{4}$$

where $\mathcal{V}_{\text{tag}}$ includes $N_{\text{tag}}$ tag nodes and $\mathcal{V}_{\text{ant}}$ contains $N_{\text{ant}}$ reader antennas, as follows

$$\mathcal{V}_{\text{tag}} = \{v_1, v_2, \ldots, v_{N_{\text{tag}}}\}, \tag{5}$$

$$\mathcal{V}_{\text{ant}} = \{v_{N_{\text{tag}}+1}, v_{N_{\text{tag}}+2}, \ldots, v_{N_{\text{tag}}+N_{\text{ant}}}\}. \tag{6}$$

The 3D coordinate for each node $n$ can be expressed by

$$c(n) = (x_n, y_n, z_n) \in \mathbb{R}^3. \tag{7}$$

The feature $x_i$ of each node $i$ can be expressed by

$$x_i = \begin{cases} \Delta\text{RSS}_i, & i = 1, \ldots, N_{\text{tag}}, \\ 0, & i = N_{\text{tag}}+1, \ldots, N_{\text{tag}}+N_{\text{ant}}, \end{cases} \tag{8}$$

where $\Delta\text{RSS}_i$ is the RSS variation from tag $i$. Specifically, the tag node RSS feature vector at each time step is

$$\mathbf{x}_{\text{tag}} = \left[\Delta\text{RSS}_1, \Delta\text{RSS}_2, \ldots, \Delta\text{RSS}_{N_{\text{tag}}}\right]^\top \in \mathbb{R}^{N_{\text{tag}}}, \tag{9}$$

where $\top$ represents transposition. Antenna nodes' RSS feature vector is set to zero, as follows

$$\mathbf{x}_{\text{ant}} = \mathbf{0} \in \mathbb{R}^{N_{\text{ant}}}. \tag{10}$$

By stacking all $N = N_{\text{tag}} + N_{\text{ant}}$ node features at each time step $t$, we can obtain a combined feature vector

$$\mathbf{x}^{(t)} = \begin{bmatrix} \mathbf{x}_{\text{tag}} \\ \mathbf{x}_{\text{ant}} \end{bmatrix} = \left[x_1^{(t)}, x_2^{(t)}, \ldots, x_N^{(t)}\right]^\top \in \mathbb{R}^N. \tag{11}$$

For the edge set $\mathcal{E}$, we define four types of edges, given by

$$\mathcal{E} = \mathcal{E}_{\text{tag-tag}} \cup \mathcal{E}_{\text{tag-ant}} \cup \mathcal{E}_{\text{ant-ant}} \cup \mathcal{E}_{\text{self}}, \tag{12}$$

where $\mathcal{E}_{\text{tag-tag}}$, $\mathcal{E}_{\text{tag-ant}}$, $\mathcal{E}_{\text{ant-ant}}$, and $\mathcal{E}_{\text{self}}$ are the sets of tag-tag grid edges, tag-antenna edges, antenna-antenna edges, and self-loop edges, respectively.

The tag-tag edge is in grid form. Each node has four neighbors in the grid, which is formulated on the $N_r \times N_c$ lattice. Each tag node can only connect to its immediate horizontal or vertical neighbors.

$$\begin{aligned} \mathcal{E}_{\text{tag-tag}} = \Big\{ \{v_{r,c},\, v_{r',c'}\} \;\Big|\; & v_{r,c},\, v_{r',c'} \in \mathcal{V}_{\text{tag}}, \\ & (r,c), (r',c') \in \{1, \ldots, N_r\} \times \{1, \ldots, N_c\}, \\ & |r - r'| + |c - c'| = 1 \Big\}, \end{aligned} \tag{13}$$

where the condition $|r - r'| + |c - c'| = 1$ is constraint to the vertical and horizontal neighbors. For example, $|r - r'| = 1, |c - c'| = 0$ means a a vertical neighbor and $|r - r'| = 0, |c - c'| = 1$ is a horizontal neighbor.

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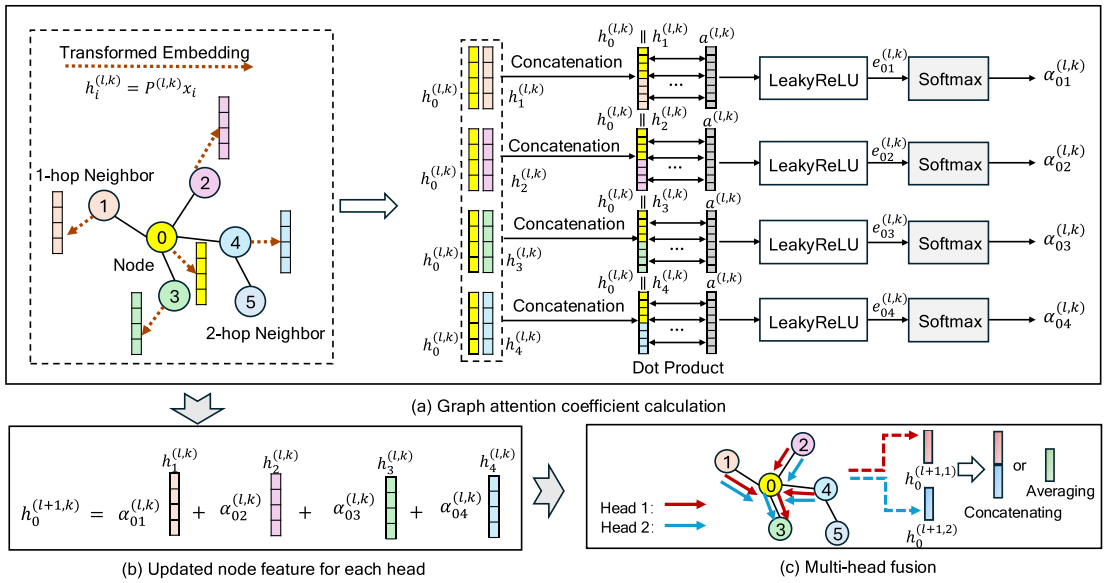



Fig. 4. Diagram of multi-head graph attention network for layer $\ell$.

The tag–antenna edge is defined as

$$\mathcal{E}_{\text{tag-ant}} = \left\{ \{v, u\} \;\middle|\; v \in \mathcal{V}_{\text{tag}}, u \in \mathcal{V}_{\text{ant}} \right\}. \tag{14}$$

The antenna–antenna edge is defined as follows

$$\mathcal{E}_{\text{ant-ant}} = \left\{ \{u_p, u_q\} \;\middle|\; u_p, u_q \in \mathcal{V}_{\text{ant}}, p \neq q \right\}. \tag{15}$$

The self-loop edge is defined as

$$\mathcal{E}_{\text{self}} = \left\{ \{x, x\} \;\middle|\; x \in \mathcal{V} \right\}. \tag{16}$$

The spatial relationships are encoded in the adjacency matrix. We consider the Euclidean distance between different nodes, and we convert it into a symmetric weighted adjacency matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$. For nodes $i$ and $j$, we use the inverse-distance weighting, as follows

$$W_{ij} = \begin{cases} \dfrac{1}{d_{ij}}, & i \neq j,\ \{i, j\} \in \mathcal{E}, \\ 1, & i = j, \\ 0, & \text{otherwise}, \end{cases} \tag{17}$$

where $d_{ij} = \|c(i) - c(j)\|_2$ is the Euclidean distance between node $i$ and node $j$.

Each tag node only has a few local connections, such as the tag neighbors or the antenna node. However, for each antenna, it needs to connect to all tag nodes, which causes the antenna nodes to have a much higher degree than the tag nodes. If we stack multiple graph layers, the antenna nodes will repeatedly sum up more signals than the tag nodes. This can cause antenna-derived features to dominate and drown out local tag-tag interactions. Therefore, to balance every node's contribution, we need to normalize the adjacency matrix, given by [31]

$$\widehat{\mathbf{W}} = \mathbf{D}^{-\frac{1}{2}} \mathbf{W} \mathbf{D}^{-\frac{1}{2}}, \tag{18}$$

where $\mathbf{D} \in \mathbb{R}^{N \times N}$ is a degree matrix, which is in diagonal form with the row-sums of the weighted adjacency matrix $\mathbf{W}$ entries, as follows

$$D_{ii} = \sum_{j=1}^{N} W_{ij}. \tag{19}$$

*2) Graph Attention Layer:* The detailed framework of GAT layer is shown in Fig. 4. The input of the GAT layer is the node feature vector $\mathbf{x}^{(t)} \in \mathbb{R}^N$ and the symmetrically normalized adjacency matrix $\widehat{\mathbf{W}}$. We employ $K$ parallel attention heads. For each head $k = 1, \ldots, K$, the initial linear transformation to project each node's feature into an $F'$-dimensional space, given by

$$\mathbf{h}_i^{(\ell,k)} = \mathbf{P}^{(\ell,k)} x_i, \quad \mathbf{P}^{(\ell,k)} \in \mathbb{R}^{F' \times 1}, \tag{20}$$

where $\mathbf{P}^{(\ell,k)}$ is the weight matrix of the GAT layer $\ell$ ($1 \leq \ell \leq L$), and $x_i$ is the feature of node $i$.

The attention coefficient between node $i$ and its neighbor $j \in \mathcal{N}(i)$, $\mathcal{N}(i) = \{j : \widehat{W}_{ij} > 0\}$ computed in head $k$ is given by [32]

$$e_{ij}^{(\ell,k)} = \text{LeakyReLU}\left( \mathbf{a}^{(\ell,k)\top} \left[ \mathbf{h}_i^{(\ell,k)} \| \mathbf{h}_j^{(\ell,k)} \right] \right), \mathbf{a}^{(\ell,k)} \in \mathbb{R}^{2F'}, \tag{21}$$

where $\mathbf{a}^{(\ell,k)} \in \mathbb{R}^{2F'}$ is a learnable attention vector, $\|$ denotes the concatenation operator, and LeakyReLU is the Leaky rectified linear unit (ReLU) activation function. We only compute the attention coefficient $e_{ij}^{(\ell,k)}$ for node $j$ in the 1hop neighbourhood $\mathcal{N}(i)$ of node $i$ according to the adjacency matrix $\widehat{\mathbf{W}}$.

To make the attention scores on the same scale for each node, we normalize them over their neighborhood using a softmax function, given by

$$\alpha_{ij}^{(\ell,k)} = \frac{\exp\left(e_{ij}^{(\ell,k)}\right)}{\sum_{m \in \mathcal{N}(i)} \exp\left(e_{im}^{(\ell,k)}\right)}. \tag{22}$$

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Then we have $\sum_{j\in\mathcal{N}(i)}\alpha_{ij}^{(\ell,k)}=1$. Using these normalized attention weights, each head can compute the updated node features for node $i$ by taking a weighted sum of neighbors' projections, given by [33]

$$\mathbf{h}_i^{(\ell+1,k)}=\sigma\left(\sum_{j\in\mathcal{N}(i)}\alpha_{ij}^{(\ell,k)}\,\mathbf{h}_j^{(\ell,k)}\right), \tag{23}$$

where $\sigma(\cdot)$ denotes the exponential linear unit (ELU) activation function. Note that $\alpha_{ij}$ does not directly capture features. It is a learnable weighting mechanism that adaptively aggregates neighbor information, thereby allowing the model to assign higher weights to more informative tag-tag and tag-antenna links during training.

After computing $K$ independent attention-head outputs $\{\mathbf{h}_i^{(\ell+1,k)}\}_{k=1}^{K}$, we need to fuse them into a single vector $\mathbf{h}_i^{(\ell+1)}$. There are two common strategies, i.e., concatenation and averaging. The concatenating function can be expressed by

$$\mathbf{h}_i^{(\ell+1)}=\Big\|_{k=1}^{K}\mathbf{h}_i^{(\ell+1,k)}. \tag{24}$$

By concatenating, we can preserve the full $K\times F'$ dimensions of all heads, which allows each head to learn and retain distinct aspects of the local neighborhood structure. For example, one head might focus on tag–tag interactions while another might emphasize tag–antenna links.

The averaging function is given by

$$\mathbf{h}_i^{(\ell+1)}=\frac{1}{K}\sum_{k=1}^{K}\mathbf{h}_i^{(\ell+1,k)}. \tag{25}$$

After averaging, we can obtain a single $F'$-dimensional vector regardless of $K$, which keeps the model size constant. This is especially useful when model capacity is limited or when a compact representation is desired.

### B. Temporal Module

The temporal module transforms the spatial embeddings obtained from the spatial module into a compact representation that captures temporal dynamics.

After processing through $L$ spatial GAT layers, we obtain a vector for each node $i$ at time step $t$, denoted by $\mathbf{h}_i^L(t)$. Stacking these embeddings across all $N$ nodes, we can obtain a matrix of spatial embeddings at time step $t$, given by

$$\mathbf{H}^{(t)}=\left[\mathbf{h}_1^{(L)}(t),\mathbf{h}_2^{(L)}(t),\ldots,\mathbf{h}_N^{(L)}(t)\right]^{\top}\in\mathbb{R}^{N\times F'}. \tag{26}$$

Then we flatten each matrix into a vector, as follows

$$\mathbf{s}_t=\mathrm{vec}\left(\mathbf{H}^{(t)}\right)\in\mathbb{R}^{F_{\text{in}}},\qquad F_{\text{in}}\triangleq NF'. \tag{27}$$

The input sequence matrix is obtained by stacking the vectors from all $T$ time steps, given by

$$\mathbf{S}=[\mathbf{s}_1,\ldots,\mathbf{s}_T]^{\top}\in\mathbb{R}^{T\times F_{\text{in}}}. \tag{28}$$

The temporal encoder $\mathcal{T}(\cdot)$ processes this sequence matrix $\mathbf{S}$ into a concise temporal representation $\mathbf{g}\in\mathbb{R}^{F_g}$, as follows

$$\mathbf{g}=\mathcal{T}(\mathbf{S};\Theta_{\mathcal{T}}), \tag{29}$$

where $\Theta_{\mathcal{T}}$ denotes the parameters of the temporal model. In this paper, we implemented the temporal encoder using three alternative approaches: LSTM, gated recurrent unit (GRU), and temporal convolutional network (TCN). The resulting embeddings are denoted as $\mathbf{g}_{\text{LSTM}}$, $\mathbf{g}_{\text{GRU}}$, $\mathbf{g}_{\text{TCN}}$, respectively. We will give a detailed description of these three temporal analysis methods.

*LSTM:* The LSTM is a general and popular method for temporal analysis, which includes an input gate $\mathbf{i}_t$, a forget gate $\mathbf{f}_t$, an output gate $\mathbf{o}_t$, and a cell candidate $\widetilde{\mathbf{c}}_t$. At time step $t$, given the input $\mathbf{s}_t\in\mathbb{R}^{F_{\text{in}}}$ and previous hidden state $\mathbf{h}_{t-1}\in\mathbb{R}^{F_h}$ and cell state $\mathbf{c}_{t-1}\in\mathbb{R}^{F_h}$, the gates and candidate are [34]

$$\mathbf{i}_t=\sigma\big(W_i\mathbf{s}_t+U_i\mathbf{h}_{t-1}+\mathbf{b}_i\big), \tag{30a}$$

$$\mathbf{f}_t=\sigma\big(W_f\mathbf{s}_t+U_f\,\mathbf{h}_{t-1}+\mathbf{b}_f\big), \tag{30b}$$

$$\widetilde{\mathbf{c}}_t=\tanh\big(W_c\mathbf{s}_t+U_c\mathbf{h}_{t-1}+\mathbf{b}_c\big), \tag{30c}$$

$$\mathbf{o}_t=\sigma\big(W_o\mathbf{s}_t+U_o\mathbf{h}_{t-1}+\mathbf{b}_o\big), \tag{30d}$$

where $\sigma(\cdot)$ is the sigmoid activation, and $\tanh(\cdot)$ is the hyperbolic tangent. Each gate weight $W_\star,\star\in\{i,f,c,o\}$ projects the current input $\mathbf{s}_t$ into the gate space, while $U_\star$ applies to the previous hidden state. The forget gate $\mathbf{f}_t$ decides which portions of the old cell state to retain. The input gate $\mathbf{i}_t$ controls how much of the new candidate $\widetilde{\mathbf{c}}_t$ enters the cell. The output gate $\mathbf{o}_t$ modulates which parts of the updated cell state are exposed as the new hidden state.

The cell state and hidden state are given by

$$\mathbf{c}_t=\mathbf{f}_t\odot\mathbf{c}_{t-1}+\mathbf{i}_t\odot\widetilde{\mathbf{c}}_t, \tag{31}$$

$$\mathbf{h}_t=\mathbf{o}_t\odot\tanh(\mathbf{c}_t). \tag{32}$$

Stacking $D$ layers yields final hidden states $\{\mathbf{h}_t^{(d)},\mathbf{c}_t^{(d)}\}_{d=1}^{D}$. The sequence embedding is

$$\mathbf{g}_{\text{LSTM}}=\mathbf{h}_T^{(D)}\in\mathbb{R}^{F_h}. \tag{33}$$

*GRU:* The GRU is simpler than LSTM with only two efficient gated computational units: an update gate and a reset gate. It controls information flow with an update gate $\mathbf{z}_t$, a reset gate $\mathbf{r}_t$, and a candidate activation $\widetilde{\mathbf{f}}_t$ at time step $t$. The update gate controls how much information from the previous hidden state will pass onto the current hidden state. Given the current input vector $\mathbf{s}_t$ and the previous hidden state $\mathbf{f}_{t-1}$, we can derive the update gates as

$$\mathbf{z}_t=\sigma\left(W_z\mathbf{s}_t+U_z\mathbf{f}_{t-1}+\mathbf{b}_z\right), \tag{34}$$

where $W_z\in\mathbb{R}^{F_f\times F_{\text{in}}}$ are the weight matrix of the current input $\mathbf{s}_t$, $U_z\in\mathbb{R}^{F_f\times F_f}$ is weight matrix of the previous state, $\mathbf{b}_z\in\mathbb{R}^{F_f}$ is the update gate bias vector, and $F_f$ is the dimension of the GRU's hidden state vector.

The reset gate determines how much of the past information should be forgotten, which is expressed by

$$\mathbf{r}_t=\sigma\left(W_r\mathbf{s}_t+U_r\mathbf{f}_{t-1}+\mathbf{b}_r\right), \tag{35}$$

where $W_r\in\mathbb{R}^{F_f\times F_{\text{in}}}$ is the weight of the reset gate, $U_r\in\mathbb{R}^{F_f\times F_h}$ is the weight matrix of reset gate, and $\mathbf{b}_r\in\mathbb{R}^{F_f}$ is the reset gate bias vector. The current memory is calculated by

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integrating the reset gate to selectively incorporate information from the previous hidden state, defined as [35]

$$\widetilde{\mathbf{f}}_t = \tanh\left(W_f\,\mathbf{s}_t + U_f\left[\mathbf{r}_t \odot \mathbf{f}_{t-1}\right] + \mathbf{b}_f\right), \tag{36}$$

$$\mathbf{f}_t = (1-\mathbf{z}_t)\odot \mathbf{f}_{t-1} + \mathbf{z}_t \odot \widetilde{\mathbf{f}}_t, \tag{37}$$

where $\tanh(\cdot)$ is the hyperbolic tangent function, $\odot$ denotes element-wise multiplication, $W_f \in \mathbb{R}^{F_f \times F_{\text{in}}}$ is the weight matrix projecting the current input $\mathbf{s}_t$ into the candidate activation, $U_f \in \mathbb{R}^{F_f \times F_h}$ is the recurrent weight matrix applied to the gated previous state $(\mathbf{r}_t \odot \mathbf{f}_{t-1})$, and $\mathbf{b}_f \in \mathbb{R}^{F_f}$ is the bias vector for the candidate activation.

Similar to the LSTM, the GRU produces the final hidden state from the last time step as its temporal embedding:

$$\mathbf{g}_{\text{GRU}} = \mathbf{h}_T^{(D)} \in \mathbb{R}^{F_f}. \tag{38}$$

*TCN:* The TCN is an efficient parallel neural architecture for sequential learning, which utilizes 1D dilated causal convolutions to capture long-range temporal dependencies. Given the input sequence $\mathbf{S} \in \mathbb{R}^{T \times F_{\text{in}}}$, the output of a single convolutional layer at time step $t$ and layer $l$ is defined as [36]

$$\mathbf{y}_t^{(l)} = \sum_{i=0}^{k-1} W_i^{(l)} \mathbf{s}_{t-d\times i}^{(l-1)} + \mathbf{b}^{(l)}, \tag{39}$$

where $k$ is the kernel size, $d$ is the dilation factor, $W_i^{(l)}$ is the convolutional kernel at position $i$ of layer $l$, and $\mathbf{b}^{(l)}$ is the bias term. The dilation factor $d$ exponentially increases with layer depth $d = 2^{l-1}$, which allows the receptive field to cover large temporal ranges.

A residual connection is employed for each layer, which improves the gradient propagation and model stability. It can be written as

$$\mathbf{s}_t^{(l)} = \sigma\left(\mathbf{y}_t^{(l)} + \mathbf{s}_t^{(l-1)}\right). \tag{40}$$

The final temporal embedding vector $\mathbf{g}$ is obtained by global average pooling over the time dimension, as follows

$$\mathbf{g}_{\text{TCN}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{s}_t^{(L_T)} \in \mathbb{R}^{F_c}. \tag{41}$$

where $L_T$ denotes the total number of TCN layers. The TCN's structure enables parallel computations, which can significantly improve training and inference efficiency compared to recurrent-based methods.

*Temporal Self-Attention:* For each temporal encoder, we also add a temporal self-attention mechanism to further refine the embedding. The output hidden states from LSTM, GRU, or TCN are projected into queries, keys, and values, followed by computing scaled dot-product attention across the temporal dimension. By averaging pooling of the attended features, we can obtain the final compact embedding, given by

$$\mathbf{g}_\star \in \mathbb{R}^{F_g}, \quad \star \in \text{LSTM, GRU, TCN}. \tag{42}$$

### C. Multi-Task Module

After the temporal module encodes the spatio-temporal information into a latent representation $\mathbf{g}$, it is used as input to two parallel prediction heads for fall detection and fall location estimation. The proposed multi-task decoder uses this shared latent representation $\mathbf{g}$ to simultaneously address these two tasks.

*1) Fall Detection:* The fall detection task aims to identify whether a fall event has occurred, which is formulated as a binary classification task, i.e., *fall* vs. *no fall*. This task is operated based on the encoded latent representation $\mathbf{g}$, which integrates spatio-temporal RSS patterns captured by GAT layers and temporal modeling. Specifically, the GAT captures the connectivity among tag and antenna nodes. The temporal layers aggregate signals over time. This joint representation reduces misclassification. Each input sequence consists of time-varying RSS values aligned with a hierarchical graph representation. We apply a single fully-connected layer to $\mathbf{g}$, followed by a sigmoid activation, given by

$$\hat{y}_{\text{fall}} = \sigma(\text{FC}_{\text{class}}(\mathbf{g})), \tag{43}$$

where $\text{FC}_{\text{class}}$ is the fully connected layer for the classification and $\hat{y}_{\text{fall}} \in [0,1]$ indicates the predicted likelihood of a fall event.

*2) Fall Location Estimation:* Once the model detects a fall event, it needs to estimate the specific location where the fall occurred. This involves localizing activity patterns within the hierarchical graph. By leveraging attention weights and node-level embeddings, our model can trace the fall event back to a particular node or region. The location estimation benefits from shared representations learned by the detection module. For instance, nodes corresponding to certain spatial positions or tag arrays can be emphasized when the network identifies a potential fall signature. Our model regresses the 3D spatial coordinates of the fall event. The predicted fall location is defined as

$$\hat{\mathbf{p}} = \text{FC}_{\text{loc}}(\mathbf{g}), \tag{44}$$

where $\text{FC}_{\text{loc}}$ is the fully connected layer for localization and $\hat{\mathbf{p}} \in \mathbb{R}^3$ represents the estimated $(x, y, z)$ spatial coordinates.

## IV. Performance Analysis

In this section, we will give a theoretical analysis of the proposed HMSTGAT method. We first formalize the joint optimization objective of the two tasks. Then we compare the multitask learning (MTL) paradigm with singletask baselines. Finally, we quantify the computational complexity of each model.

### A. Unified Multi-Task Objective

Given a batch of $Q$ sequences, let $y_q \in \{0,1\}$ denote the ground-truth fall label and $\mathbf{p}_q \in \mathbb{R}^3$ the Cartesian coordinates of the fall position for the $q$-th sequence. The network predicts $\hat{y}_q$ and $\hat{\mathbf{p}}_q$. We use the binary cross-entropy to represent the loss function for fall detection and the root mean squared error (RMSE) to evaluate the localization estimation loss function, respectively, expressed by

$$\mathcal{L}_{\text{fall}} = -\left[y_q \log \hat{y}_q + (1-y_q)\log(1-\hat{y}_q)\right], \tag{45}$$

$$\mathcal{L}_{\text{loc}} = \sqrt{\frac{1}{Q}\sum_{q=1}^{Q} \|\mathbf{p}_q - \hat{\mathbf{p}}_q\|_2^2}. \tag{46}$$

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TABLE I
COMPLEXITY ANALYSIS PER SEQUENCE

| Component | Time | Memory |
|---|---|---|
| **Spatial Module** | | |
| GAT ($L$ layers) | $\mathcal{O}\big(L(NF_s^2H + \lvert\mathcal{E}\rvert F_sH)\big)$ | $\mathcal{O}(NF_sH)$ |
| GCN ($L$ layers) | $\mathcal{O}\big(L(NF_s^2 + \lvert\mathcal{E}\rvert F_s)\big)$ | $\mathcal{O}(NF_s)$ |
| **Temporal Module** | | |
| LSTM | $\mathcal{O}(4TF_t^2)$ | $\mathcal{O}(TF_t)$ |
| GRU | $\mathcal{O}(3TF_t^2)$ | $\mathcal{O}(TF_t)$ |
| TCN | $\mathcal{O}(L_TK_sTF_t^2)$ | $\mathcal{O}(TF_t)$ |
| **Detection Heads** | | |
| Classification | $\mathcal{O}(F_t)$ | $\mathcal{O}(F_t)$ |
| Localization | $\mathcal{O}(3F_t)$ | $\mathcal{O}(3F_t)$ |

TABLE II
RFID SYSTEM PARAMETERS

| Parameter | Value |
|---|---|
| Reader model | ThingMagic M7e-MEGA |
| Reader operating frequency | 905 MHz |
| Transmit power | 31.5 dBm |
| Reader antennas | 2× MT-242025/TRH/A (865-956 MHz) |
| Antenna gain | 7.5 dBi |
| 3-dB elevation beamwidth | 72° |
| Tag type | Passive UHF RAIN RFID |
| Tag dimensions | 9.5 cm×3 cm |
| Deployment pattern | $4 \times 5$ grid on wall |

Then, the total loss is defined by a combination of classification loss and localization regression loss, given as [37]

$$\mathcal{L} = \lambda_{\text{fall}}\mathcal{L}_{\text{fall}}(y_q, \hat{y}_q) + \lambda_{\text{loc}}\mathcal{L}_{\text{loc}}(\mathbf{p}_q, \hat{\mathbf{p}}_q), \tag{47}$$

where $\lambda_{\text{fall}}$ and $\lambda_{\text{loc}}$ are the controlled hyperparameters of fall detection and fall localization estimation function, respectively.

### B. *Single-Task* vs. *Multi-Task Learning*

To quantify the benefit of jointly learning detection and localization, we implemented two single-task baselines:

1) *HM-STGAT*$_{\text{Det}}$*:* utilizes the identical architecture, retaining only the detection head while removing the localization head and its corresponding loss function, i.e., $\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$.
2) *HM-STGAT*$_{\text{Loc}}$*:* retains only the localization head and its loss function, removing the detection head, i.e., $\lambda_{\text{fall}} = 0$, $\lambda_{\text{loc}} = 1$.

For each baseline, we can use the temporal module only, the spatial module only, or the integrated spatio-temporal module to evaluate performance.

### C. *Complexity Analysis*

We analyze the computational cost of each component in our model, as shown in Table I. For the spatial module, the GAT has a time complexity of $\mathcal{O}(L(NF_s^2H + |\mathcal{E}|F_sH))$ and memory complexity of $\mathcal{O}(NF_sH)$, where $L$ is the number of layers, $N$ is the number of nodes, $F_s$ is the feature dimension, $H$ is the number of attention heads, and $|\mathcal{E}|$ is the number of edges.

TABLE III
FALL LOCATION COORDINATES ($z = 0$)

| ID | Coordinate | ID | Coordinate | ID | Coordinate |
|---|---|---|---|---|---|
| $F_1$ | (−80, 37.5, 0) | $F_2$ | (−40, 37.5, 0) | $F_3$ | (0, 37.5, 0) |
| $F_4$ | (40, 37.5, 0) | $F_5$ | (80, 37.5, 0) | $F_6$ | (−80, 75, 0) |
| $F_7$ | (−40, 75, 0) | $F_8$ | (0, 75, 0) | $F_9$ | (40, 75, 0) |
| $F_{10}$ | (80, 75, 0) | $F_{11}$ | (−80, 112.5, 0) | $F_{12}$ | (−40, 112.5, 0) |
| $F_{13}$ | (0, 112.5, 0) | $F_{14}$ | (40, 112.5, 0) | $F_{15}$ | (80, 112.5, 0) |

The graph convolutional network (GCN) is more efficient, with time and memory costs of $\mathcal{O}(L(NF_s^2 + |\mathcal{E}|F_s))$ and $\mathcal{O}(NF_s)$, respectively.

For the temporal module, the LSTM has a slightly higher computational cost than GRU. The TCN has a complexity of $\mathcal{O}(L_TK_sTF_t^2)$ in time, with $L_T$ denoting the number of convolutional levels and $K_s$ the kernel size, and $\mathcal{O}(TF_t)$ in memory.

Finally, the detection heads for classification and localization have the $\mathcal{O}(F_t)$ and $\mathcal{O}(3F_t)$ time and memory complexity, respectively.

## V. EXPERIMENT SETUP

This experiment was conducted in a controlled indoor laboratory environment with a fixed layout. Fig. 5(a) is the overview scene of the experiment setup with a pair of patch antennas placed on a low table, both wired into an RFID reader unit. The reader is connected to a laptop that can automatically obtain the RSS values from each passive RFID tag in dBm through the Universal Reader Assistant software. Facing the antennas is the passive tag wall, where the tags are mounted in a $4 \times 5$ grid. Fig. 5(b) is the passive tag used in the experiment, each approximately 9.5 cm in length and 3 cm in width. Fig. 5(c) is a wider view of the full tag array that is arranged in a regular pattern, forming a two-dimensional sensing surface. Fig. 5(d) is the detailed shot of the patched antennas with a gain of 7.5 dBi and a 3 dB elevation beamwidth of 72° used to illuminate the tag grid. They are mounted side by side on the table and directed to the tag plane. The small white markers on the carpeted floor in front of the tag wall in Fig. 5(e) indicate the precise spots where the human is instructed to fall or other movements during each trial. All relevant hardware and radio parameters are summarized in Table II.

We establish a 3D coordinate system for the tags and reader antennas, as shown in Fig. 6. The wall is defined as the $(x, z)$ plane, and the perpendicular distance from the wall to the reader is the $y$-axis. Then the floor corresponds to $z = 0$. The third tag column is centered at $x = 0$, and the tag wall plane is $y = 0$. All coordinates are in centimeters. The coordinates of each tag are as follows

The coordinates of reader antenna 1 and 2 are $(20, 150, 70)$ and $(-20, 150, 70)$, respectively. We also defined 15 potential fall-location markers on the floor, as shown in Table III.

As shown in Table IV, the dataset for each antenna consists of two main categories: fall scenarios and non-fall scenarios, with 100 sequences in each category. With two antennas used in this experiment, the total number of training sequences is 400. It is important to note that collecting large training datasets is

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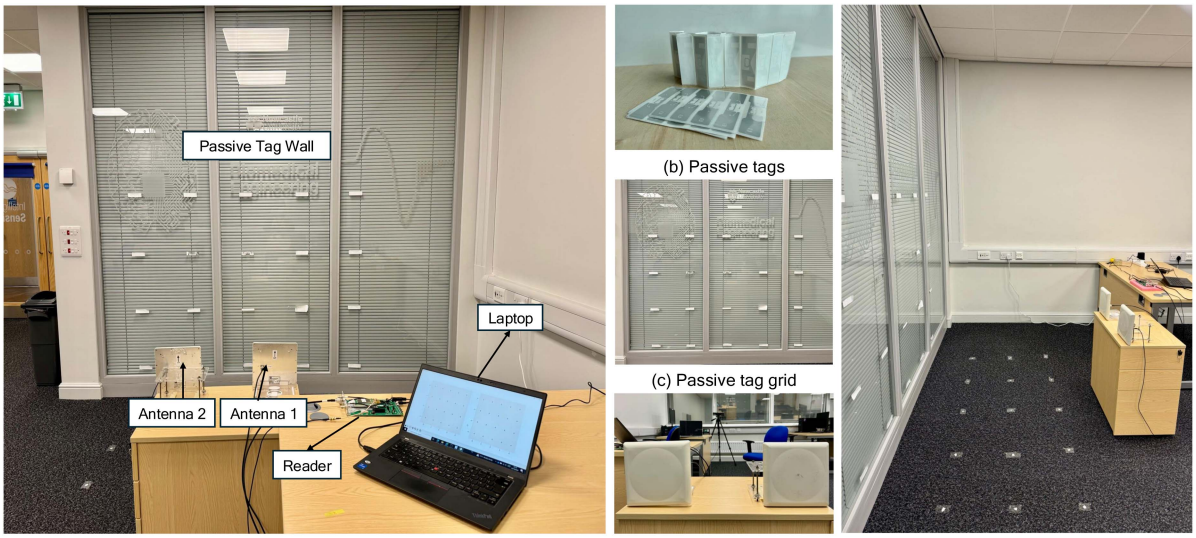



Fig. 5. Experimental setup for RFIDbased fall detection: passive RFID tag grid, reader antennas, and fall location markers.

TABLE IV
CONFIGURATION OF FALL AND NON-FALL SCENARIO DATASETS

| Category | Scenario Type | Number of Sequences | Description | Time Steps |
|---|---|---|---|---|
| Fall Scenarios | Static Fall Scenario | 15 | Forward falls at 15 distinct positions | 20 |
| | | 15 | Backward falls at 15 distinct positions | 20 |
| | | 15 | Sideway falls at 15 distinct positions | 20 |
| | Fall-and-Recover Scenario | 15 | Forward fall followed by standing | 20 |
| | | 15 | Backward fall followed by standing | 20 |
| | | 15 | Sideway fall followed by standing | 20 |
| | Additional Fall Scenarios | 5 | Slow-motion fall events | 30 |
| | | 5 | Sudden fall events | 30 |
| Non-Fall Scenarios | Standing | 15 | Standing parallel to the wall | 20 |
| | | 15 | Standing perpendicular to the wall | 20 |
| | Walking | 20 | Walking parallel to the wall | 20 |
| | | 20 | Walking perpendicular to the wall | 20 |
| | | 10 | Walking along a diagonal trajectory | 20 |
| | | 10 | Walking in a Z-shaped trajectory | 30 |
| | | 10 | Walking in an S-shaped trajectory | 30 |

particularly challenging, as there is currently no effective publicly available dataset for RFID-based scenarios. This limitation motivates this work, which aims to develop and train a novel and efficient model capable of achieving reliable detection accuracy using a relatively small dataset.

The fall scenarios include three distinct types. The first type, static fall scenario, consists of 45 sequences lasting 20 time steps each. These sequences capture falls occurring at 15 distinct positions with forward, backward, and sideways orientations. The individual remains stationary after falling. The second type, fall-and-recover scenario, includes another set of 45 sequences, also lasting 20 time steps each. In these cases, the individual falls at the same 15 distinct positions and orientations but stands up afterward. To further diversify the scenarios, the third type includes an additional 10 sequences lasting 30 time steps each, consisting of 5 slow-motion falls and 5 sudden falls.

The non-fall scenarios are divided into standing still and walking activities. The standing still scenario contains 30 sequences, each lasting 20 time steps, where the individual stands at 15 different locations facing two orientations: parallel and perpendicular to the wall. This captures comprehensive RSS variations due to orientation changes relative to the antennas. The walking scenarios consist of 70 sequences that cover various movement patterns, including 20 sequences walking parallel to the wall and 20 sequences walking perpendicular to the wall, each lasting 20 time steps. Additionally, there are 10 sequences of diagonal walking lasting 20 time steps each, and 10 sequences

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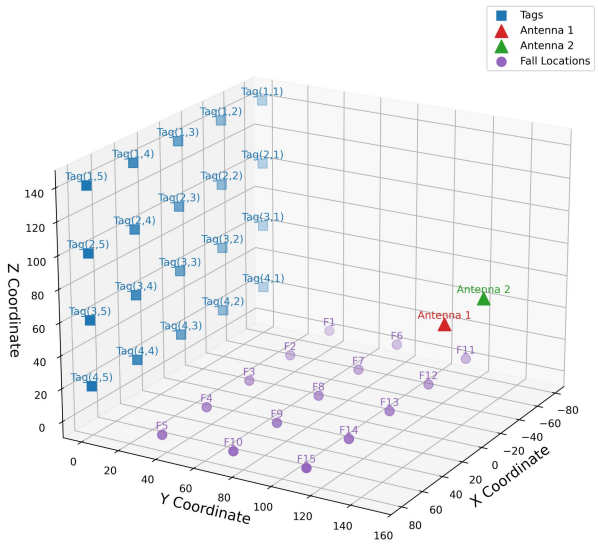



Fig. 6. 3D layout of tags, antennas, and fall locations.

each of Z-shaped and S-shaped walking patterns lasting 30 time steps each.

To evaluate the model's robustness from different spatial perspectives, the dataset is categorized into the antenna 1 dataset, the antenna 2 dataset, and the combined dataset. Antenna 1 and antenna 2 datasets each contain 200 sequences, evenly distributed between fall and non-fall scenarios. The combined dataset merges data from both antennas, resulting in 400 sequences in total. Each dataset is divided into training and test subsets, with 80% of the data used for training and 20% for testing. All splits are performed at the sequence level, with no overlap of individual sequences between training and test sets. Given the controlled experimental setup with repeated trials at predefined locations, location-level separation was not explicitly imposed. A fixed random seed was used for data splitting and model initialization to ensure reproducibility.

To standardize the data, we use the min-max normalization method to scale all $\Delta RSS$ values to a range between 0 and 1, given by

$$X_{\text{normalized}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}. \tag{49}$$

After normalization, all data are in a consistent range, which is helpful for the subsequent data analysis.

## VI. Measurement Results

For all experiments, we consider a deployment with a total of $N = 22$ nodes, which includes 20 tag nodes and 2 antenna nodes. The hierarchical graph contains $|\mathcal{E}| = 168$ edges, incorporating spatial connectivity among tags as well as tag-antenna and antenna-antenna links. The spatial feature dimension is $F_s = 32$, each graph convolutional layer uses $H = 2$ attention heads, and the temporal hidden size is $F_t = 32$. The input sequences are standardized to a fixed length of $T = 30$ time steps. Sequences with original length $T = 20$ are zero-padded at the end to reach length 30, while sequences with $T = 30$ require no padding. To prevent padded values from introducing artificial temporal patterns, padding positions are masked during temporal modeling.

The hierarchical STGAT is configured with $L = 2$ spatial layers and $L_t = 3$ temporal layers when using TCN as the temporal backbone, and the convolution kernel size is set to $K_s = 3$. All experiments use a batch size of 32. Models are trained for up to 300 epochs using the AdamW optimizer, with early stopping based on validation accuracy.

Fig. 7 shows the fall detection accuracy of our proposed HM-STGAT model under five different loss weight settings compared with standard sequence models and graph models on the validation set over 300 epochs. The HM-STGAT of a single fall detection task, i.e. $\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$, converges the fastest and achieves the highest accuracy of about 97%. However, for the multi-task fall detection, when the localization loss weight of the proposed HM-STGAT is increased, the fall detection performance degrades. Comparing the green and blue curves shows that increasing the fall detection loss weight improves final accuracy by 12%. However, comparing the orange and blue curves reveals that prioritizing localization loss decreases fall detection accuracy by 10%. These observations demonstrate a trade-off in the multi-task settings that emphasizes the fall detection loss helps mitigate performance degradation, but does not fully match the performance of the single-task models. Moreover, assigning equal weights to both tasks also leads to suboptimal fall detection accuracy. Among the single-task methods, the temporal-based fall detection method (LSTM, GRU, TCN) performs better than the graph-based method (GCN, GAT), which indicates the superior ability of temporal convolutional and recurrent architectures to capture the rapid signal changes of falls.

Fig. 8 summarizes the peak fall detection accuracy achieved by each evaluated model. The proposed HM-STGAT model with $\lambda_{\text{fall}} = 1$ and $\lambda_{\text{loc}} = 0$ demonstrates the highest fall-detection accuracy of 97.5%. This result indicates a significant improvement over traditional temporal-based approaches, such as TCN, LSTM, and GRU. The performance advantage of HM-STGAT proves the effectiveness of incorporating both spatial and temporal modules in our method. Among the graph-only baselines, GAT and GCN reach fall detection accuracies of 78.8% and 72.5%, respectively, which are lower than the temporal and spatio-temporal models. This gap demonstrates that while graph-based models can capture spatial relationships between tag and antenna nodes, they may not fully exploit temporal patterns for fall detection.

Additionally, we observe that variations of the HM-STGAT model with different combinations of $\lambda_{\text{fall}}$ and $\lambda_{\text{loc}}$ result in a noticeable drop in accuracy. For instance, when increasing $\lambda_{\text{loc}}$ to emphasize the location loss term, i.e., $\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0.5$ or $\lambda_{\text{fall}} = 0.5, \lambda_{\text{loc}} = 1$, the fall-detection accuracy decreases to 70.0% and 65.0%, respectively. This observation suggests that simultaneously optimizing for both fall detection and fall localization during training may compromise the model's effectiveness for single-task scenarios (48) shown at the bottom of next page.

Fig. 9 illustrates the fall localization RMSE on the test dataset over 300 training epochs for different models. It is evident that

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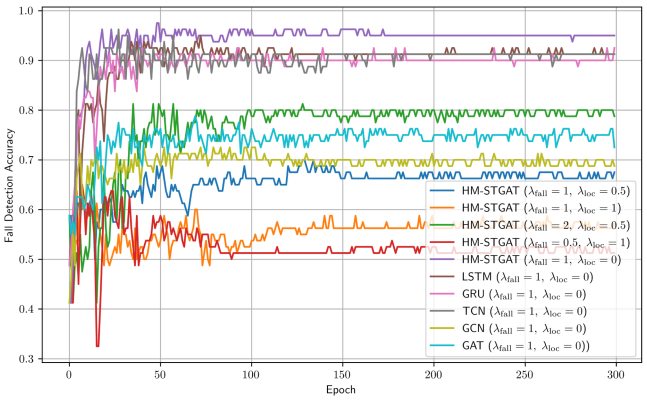



Fig. 7. Comparison of fall detection accuracy with different models and loss weight settings.

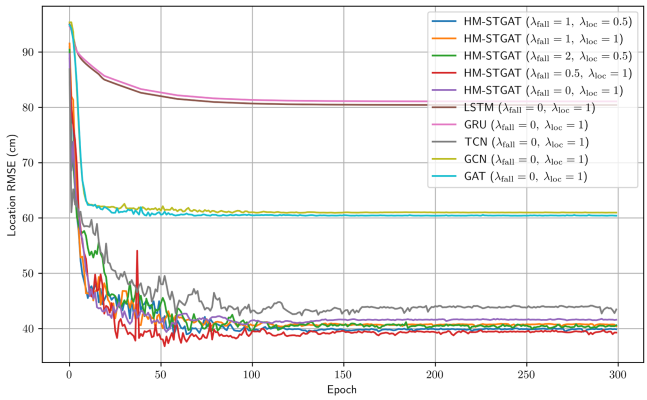



Fig. 9. Fall localization RMSE over training epochs for different models.

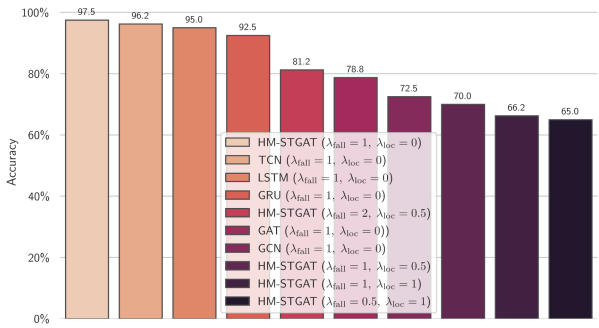



Fig. 8. Best fall-detection accuracy across different models.

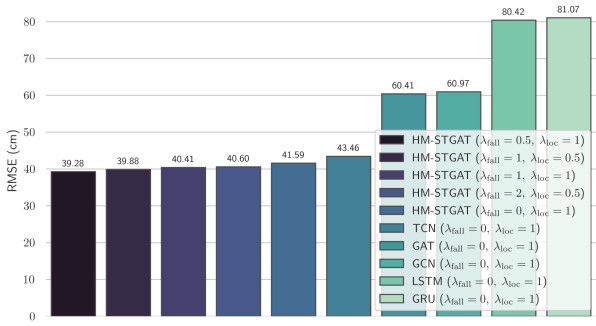



Fig. 10. Best fall localization RMSE of different models.

the HM-STGAT model with $\lambda_{\text{fall}} = 0.5$ and $\lambda_{\text{loc}} = 1$ achieves the lowest RMSE among all evaluated models. This demonstrates the advantage of assigning a higher weight to the location loss during training, which encourages the model to prioritize precise localization. In comparison, RNN-based temporal models such as LSTM and GRU exhibit higher RMSE, which suggests that relying solely on temporal architectures without using spatial relationships is less effective for fall localization tasks. Graph-based approaches, such as GCN and GAT, show intermediate performance. While these methods can capture spatial dependencies, they do not exploit the temporal information that limits their localization accuracy compared with spatio-temporal models. Interestingly, we find that the single-task HM-STGAT trained only for localization, i.e., $\lambda_{\text{fall}} = 0$ and $\lambda_{\text{loc}} = 1$, does not perform as well as its multi-task counterpart. This is because fall events and fall localization are correlated tasks, and a single-task model may miss these connections. By optimizing for both objectives simultaneously, the multi-task HM-STGAT is encouraged to capture and utilize shared representations and patterns that benefit both tasks.

Fig. 10 presents the best fall localization RMSE achieved by different models. Among all methods, the HM-STGAT model with $\lambda_{\text{fall}} = 0.5$ and $\lambda_{\text{loc}} = 1$ achieves the lowest RMSE of 39.28 cm, indicating the most accurate fall localization performance. Other HM-STGAT with different loss weights also achieve low RMSE values ranging from 39.28 cm to 41.59 cm, which demonstrates the effectiveness and robustness of the spatio-temporal approach for localization. In contrast, temporal-only models such as LSTM and GRU have higher RMSE values, at 80.42 cm and 81.07 cm, respectively, which indicates that sequence modeling alone is insufficient for precise fall localization. Graph-based models achieve intermediate performance, with RMSEs of 60.41 cm and 60.97 cm, respectively, outperforming temporal models but below the HM-STGAT models.

Fig. 11 shows the weighted training loss curves of different models over 300 epochs. As seen in the figure, all models exhibit a rapid decrease in loss during the initial training phase. Among

$$\begin{bmatrix} (-80,0,140) & (-40,0,140) & (0,0,140) & (40,0,140) & (80,0,140) \\ (-80,0,100) & (-40,0,100) & (0,0,100) & (40,0,100) & (80,0,100) \\ (-80,0,60) & (-40,0,60) & (0,0,60) & (40,0,60) & (80,0,60) \\ (-80,0,20) & (-40,0,20) & (0,0,20) & (40,0,20) & (80,0,20) \end{bmatrix}. \tag{48}$$

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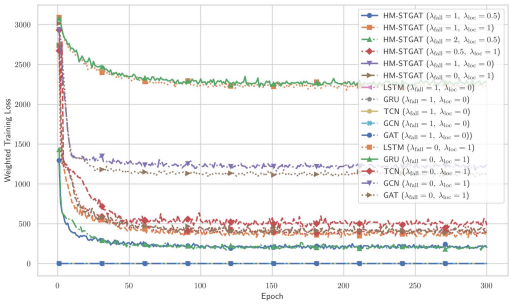



Fig. 11. Training loss curves for all models.

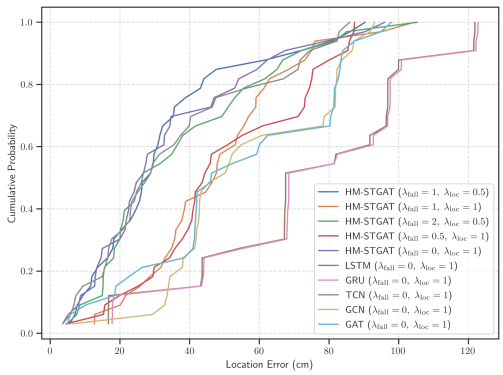



Fig. 12. CDF of fall location estimation errors for different models.

the models, LSTM and GRU for the single-task of fall localization exhibit the highest weighted training losses. The GCN and GAT models demonstrate intermediate training losses for fall localization, which performs better than purely temporal models. In contrast, the LSTM, GRU, and TCN models trained for the single-task of fall detection achieve the lowest training losses. This observation suggests that fall detection is a relatively easy task for these temporal models. The HM-STGAT models also achieve low and stable training losses across various settings, which shows its effectiveness for complex tasks that require simultaneous detection and localization.

Fig. 12 presents the cumulative distribution functions (CDFs) of fall location estimation errors for different models. The CDF curves display the proportion of test cases for which the location estimation error falls below a given value. The steeper and left-shifted curves mean more accurate predictions and a higher proportion of small errors. From the figure, it is evident that the HM-STGAT models achieve higher cumulative probabilities at lower error thresholds. This indicates that the proposed HM-STGAT models can estimate the fall location more accurately for a greater portion of the test samples.

Fig. 13 displays the confusion matrix for the top-three fall detection models, i.e., HM-STGAT, TCN, and LSTM, each trained with $\lambda_{\text{fall}} = 1$ and $\lambda_{\text{loc}} = 0$. Each matrix shows the number of true positives, true negatives, false positives, and false negatives on the test set. The HM-STGAT model achieves the highest accuracy, with 46 true negatives, 32 true positives, and only one misclassification in each of the false positive and false negative categories. These results indicate that HM-STGAT provides the most balanced and reliable fall detection.

TABLE V
CLASSIFICATION METRICS FOR DIFFERENT MODELS

| Method | Accuracy | Precision | Recall | $F_1$ Score |
|---|---|---|---|---|
| HM-STGAT ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0.5$) | 0.700 | 0.600 | 0.818 | 0.692 |
| HM-STGAT ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 1.0$) | 0.662 | 0.550 | 1.000 | 0.710 |
| HM-STGAT ($\lambda_{\text{fall}} = 2, \lambda_{\text{loc}} = 0.5$) | 0.813 | 1.000 | 0.545 | 0.706 |
| HM-STGAT ($\lambda_{\text{fall}} = 0.5, \lambda_{\text{loc}} = 1.0$) | 0.650 | 0.632 | 0.364 | 0.462 |
| HM-STGAT ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$) | 0.975 | 0.970 | 0.970 | 0.970 |
| LSTM ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$) | 0.950 | 0.939 | 0.939 | 0.939 |
| GRU ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$) | 0.925 | 0.935 | 0.879 | 0.906 |
| TCN ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$) | 0.963 | 0.941 | 0.970 | 0.955 |
| GCN ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$) | 0.725 | 0.608 | 0.939 | 0.738 |
| GAT ($\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$) | 0.788 | 0.674 | 0.939 | 0.785 |

TABLE VI
COMPUTATIONAL COST SUMMARY FOR EACH MODEL

| Model | Time (s) | Peak Memory Usage (MB) |
|---|---|---|
| HM-STGAT ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 0.5$) | 444.3 | 568.7 |
| HM-STGAT ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 1.0$) | 471.1 | 569.5 |
| HM-STGAT ($\lambda_{\text{fall}} = 2,\ \lambda_{\text{loc}} = 0.5$) | 483.8 | 567.5 |
| HM-STGAT ($\lambda_{\text{fall}} = 0.5,\ \lambda_{\text{loc}} = 1.0$) | 514.6 | 598.6 |
| LSTM ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 0$) | 219.8 | 450.3 |
| GRU ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 0$) | 238.1 | 325.7 |
| TCN ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 0$) | 226.1 | 432.7 |
| GCN ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 0$) | 226.3 | 534.7 |
| GAT ($\lambda_{\text{fall}} = 1,\ \lambda_{\text{loc}} = 0$) | 459.3 | 413.8 |
| LSTM ($\lambda_{\text{fall}} = 0,\ \lambda_{\text{loc}} = 1$) | 161.6 | 489.9 |
| GRU ($\lambda_{\text{fall}} = 0,\ \lambda_{\text{loc}} = 1$) | 193.5 | 484.2 |
| TCN ($\lambda_{\text{fall}} = 0,\ \lambda_{\text{loc}} = 1$) | 206.0 | 522.4 |
| GCN ($\lambda_{\text{fall}} = 0,\ \lambda_{\text{loc}} = 1$) | 211.8 | 420.2 |
| GAT ($\lambda_{\text{fall}} = 0,\ \lambda_{\text{loc}} = 1$) | 575.3 | 474.1 |

Table V reports the classification performance metrics in accuracy, precision, recall, and $F_1$ score for various fall detection models. The results show that the proposed HM-STGAT model with $\lambda_{\text{fall}} = 1, \lambda_{\text{loc}} = 0$ achieves the highest overall performance, with an accuracy, precision, recall, and $F_1$ score all reaching 0.97, indicating excellent and balanced fall detection capability.

As shown in Table VI, we present the CPU processing time and peak memory consumption for all evaluated models on both the fall detection and localization subtasks. For the HM-STGAT models, both the processing time and memory usage are generally higher compared to the baseline models. In contrast, the temporal models and graph-based models exhibit lower computational requirements. Among these, GRU and TCN are the most resource-efficient. GAT shows relatively high processing time, especially for the fall localization task. These results indicate a clear trade-off between model complexity and resource usage. The HM-STGAT models have superior detection and localization performance at the cost of increased computational overhead. In our prototype, the system processes RSS measurements in a short window of length $T$. Once a window is collected, HM-STGAT performs a single forward pass to output the fall decision. Therefore, the end-to-end response time mainly consists of RSS sampling time and model inference time. Since inference only involves one forward pass, its computational overhead is small compared with RSS sampling. Therefore, the system can enable near real-time continuous fall monitoring once the window is available.

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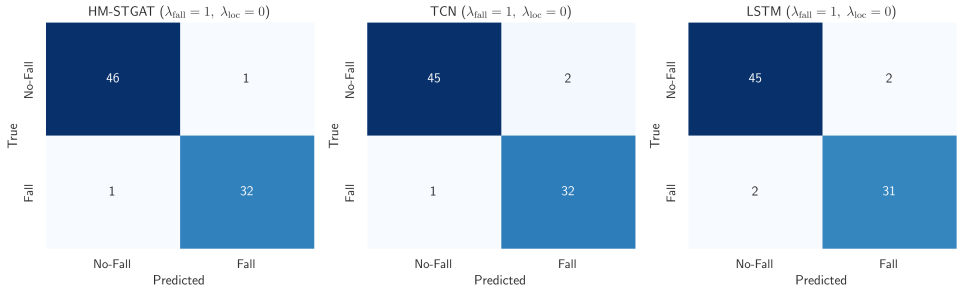



Fig. 13. Confusion matrix of top-three fall detectors.

## VII. Conclusion

In this paper, we presented a novel ambient IoT-enabled human motion detection system for healthcare monitoring, focusing on robust fall detection and localization. We proposed an advanced framework HM-STGAT that integrates spatial dependencies among passive tags and temporal dynamics of human activities. This unified approach simultaneously addresses fall event detection and precise localization, which significantly outperforms conventional temporal-only and spatial-only methods. Experimental results from a real-world indoor laboratory validated that the HM-STGAT achieved state-of-the-art performance, with high accuracy in fall detection 97.5%, and precise fall localization RMSE of 39.28 cm. Future work will explore more diverse environments and further improve the energy efficiency and privacy preservation for practical deployment in healthcare monitoring systems.

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